

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- AI Quality Engineering and Test Automation professional with over **6+ years** of experience validating enterprise applications, APIs, data pipelines, workflow-based systems, and AI-enabled solutions across regulated business environments.
- Hands-on experience testing **LLM applications, AI assistants, generative AI solutions, intelligent agents, and Retrieval-Augmented Generation (RAG) systems**, with focus on accuracy, relevance, consistency, groundedness, safety, latency, and task completion.
- Experienced designing **AI evaluation datasets, golden datasets, synthetic datasets, benchmark scenarios, adversarial cases, prompt tests, and expected outcomes** for model-driven workflows and end-user interactions.
- Strong experience validating RAG pipelines across **document ingestion, chunking, embeddings, retrieval quality, prompt behavior, response quality, citation/source alignment, and orchestration workflows**.
- Proficient with **Python, SQL, REST APIs, JSON, Playwright, Postman, automation frameworks, and data-validation utilities**, supporting functional, integration, regression, end-to-end, performance, and UAT-focused quality validation.
- Experienced identifying **hallucinations, model failure modes, bias concerns, retrieval failures, workflow breakdowns, privacy/security risks, and integration defects**, supported by RCA, risk assessment, traceability, and corrective-action recommendations.
- Strong Quality Engineering leadership across Agile teams, partnering with AI Engineers, Software Engineers, Product Owners, Business Analysts, and stakeholders on acceptance criteria, AI governance, observability, release readiness, and continuous improvement.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **AI/LLM Quality Engineering:** LLM Testing, Generative AI, AI Assistants, Intelligent Agents, RAG, Prompt Testing, LLM Evaluation, Hallucination Testing, Groundedness, Relevance, Consistency, Safety, Task Completion.
- **AI Evaluation & Test Data:** Golden Datasets, Synthetic Datasets, Evaluation Datasets, Benchmark Scenarios, Adversarial Testing, Acceptance Criteria, Rubric-Based Evaluation, Semantic Similarity, Embeddings, Automated Evaluation Frameworks.
- **RAG & Retrieval Validation:** Document Ingestion, Chunking, Embeddings, Vector Search, Retrieval Quality, Context Relevance, Citation/Source Alignment, Prompt Behavior, Response Quality, Orchestration Workflows.
- **Programming & Data:** Python, SQL, Java, C#, TypeScript, JavaScript, REST APIs, JSON, SQL Server, PostgreSQL, Data Preprocessing, Feature Validation, Data Quality, Data-Flow Testing.
- **Testing & Automation:** Playwright, Selenium WebDriver, Pytest, NUnit, TestNG, Postman, Functional, Integration, Regression, End-to-End, Performance, Usability, UAT, API and Data-Pipeline Testing.
- **Cloud & AI Platforms:** Azure AI, Azure OpenAI, AWS Bedrock, Amazon Web Services, Google Vertex AI, OpenAI APIs, CI/CD, Azure DevOps, Jenkins, Git, Model Monitoring, AI Observability, Drift Detection.
- **Governance & Delivery:** Responsible AI, AI Governance, Model Risk, Privacy, Security, Compliance Controls, NIST AI RMF Concepts, Traceability, Defect Management, RCA, Quality Metrics, Release Readiness, Agile/Scrum.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

Charlotte, NC, USA

- Define and execute quality strategies for AI-enabled enterprise capabilities using **Python, C#, Playwright, REST APIs, SQL, and automated evaluation utilities**, covering functional, integration, regression, usability, performance, and end-to-end testing.
- Validate **LLM assistants and RAG workflows** for accuracy, relevance, consistency, groundedness, safety, latency, instruction adherence, and end-to-end task completion across complex business scenarios and user interactions.
- Test RAG pipelines across document ingestion, preprocessing, chunking, embeddings, retrieval, prompt construction, response generation, and citation/source alignment while identifying context loss, irrelevant retrieval, and hallucination risks.
- Develop **golden datasets, synthetic test data, benchmark scenarios, prompt suites, adversarial cases, and expected outcomes** to provide repeatable evaluation of model behavior and AI regression across releases.
- Perform data and integration validation using Python, SQL Server, REST/JSON APIs, and reconciliation checks to verify preprocessing, feature/data quality, AI workflow dependencies, backend processing, and downstream system consistency.
- Automate AI evaluations and regression checks within **Azure DevOps/TeamCity CI/CD**, while monitoring response quality, latency, operational metrics, user feedback, and failure trends to identify model or workflow degradation.
- Partner with AI Engineers, developers, Product Owners, and business stakeholders on acceptance criteria, quality risks, traceability, defect RCA, responsible AI controls, release evidence, and recommendations for production readiness.

Environments: Python, C#, Playwright, REST APIs, JSON, SQL Server, RAG, LLM Evaluation, Embeddings, Vector Search, Golden/Synthetic Datasets, Hallucination/Groundedness Testing, Azure OpenAI, Azure DevOps, CI/CD, AI Observability, Responsible AI, RCA, Agile/Scrum.

SDET / QA Automation Engineer

Client: CVS

Dec 2023 – Sept 2024

Charlotte, NC, USA

- Performed quality engineering across regulated healthcare applications using **Python, Java, TypeScript, Playwright, Selenium WebDriver, RESTAssured, SQL, and Postman**, covering APIs, workflows, integrations, and data-driven application behavior.
- Validated AI/LLM-enabled workflows using structured prompts, business scenarios, expected-response criteria, and evaluation datasets to assess accuracy, relevance, completeness, consistency, groundedness, and operational usability.
- Created positive, negative, boundary, and adversarial scenarios to identify **hallucinations, unsafe responses, data leakage, inconsistent outputs, context failures, edge cases, and model-driven workflow breakdowns**.
- Performed API, data-pipeline, and backend validation using REST/JSON, SQL, reconciliation queries, and test-data controls to verify data preprocessing, transformations, system dependencies, and cross-application consistency.
- Built automated regression coverage for deterministic application behavior and AI-driven responses, integrating quality checks with **Jenkins, GitLab/GitHub, Git, and CI/CD pipelines** for repeatable release validation.
- Analyzed application logs, API traces, SQL data, Splunk telemetry, AI outputs, and failed evaluations to perform root-cause analysis and distinguish model issues from data, integration, environment, or application defects.
- Collaborated with engineers, Product stakeholders, analysts, and QA teams to establish acceptance criteria, communicate quality metrics and risks, validate fixes, and support release-readiness decisions in a regulated environment.

Environments: Python, Java, TypeScript, Playwright, Selenium WebDriver, RESTAssured, Postman, JSON, SQL, AI/LLM Testing, Evaluation Datasets, Hallucination Testing, Jenkins, GitLab/GitHub, Splunk, CI/CD, Data Validation, Healthcare Compliance, Agile/Scrum.

Software Engineer

Infor

Sept 2021 – Dec 2022

Hyderabad, India

- Developed and executed automated quality solutions using **Python, Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL** across enterprise applications, APIs, workflow systems, and integrated data services.

- Created functional, integration, regression, end-to-end, and performance scenarios from business requirements and acceptance criteria while identifying edge cases, integration dependencies, and measurable quality risks.
- Built Python-based test utilities for API validation, structured-data comparison, test-data generation, response analysis, and repeatable evaluation of complex backend and rules-driven application workflows.
- Performed data-pipeline and backend testing through SQL, REST/SOAP APIs, JSON/XML, source-to-target reconciliation, preprocessing validation, transformation checks, and cross-system data-flow verification.
- Integrated automated test suites with **Jenkins, Git, Maven, and CI/CD pipelines**, supporting automated regression, repeatable validation, build-quality checks, and faster identification of integration or data defects.
- Investigated failures using application logs, API responses, SQL results, execution evidence, and debugging techniques while performing root-cause analysis and expanding regression coverage around recurring quality risks.
- Worked with developers, analysts, QA engineers, and product stakeholders to define test standards, maintain traceability, document quality evidence, review risks, and continuously improve enterprise Quality Engineering practices.

Environments: Python, Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, REST/SOAP APIs, JSON/XML, SQL, Data Pipelines, Data Reconciliation, Jenkins, Git, Maven, CI/CD, Quality Metrics, RCA, Agile/Scrum.

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Executed comprehensive quality validation across enterprise web, API, data, and workflow-based systems using **Java, Python, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL**.
- Developed functional, integration, regression, end-to-end, performance, and UAT-focused test scenarios from requirements, business workflows, acceptance criteria, and integration specifications across complex enterprise applications.
- Validated REST/SOAP APIs, JSON/XML payloads, backend processes, and database records using Postman, SQL, and reconciliation techniques to ensure correct data flow, transformations, error handling, and system integration behavior.
- Created structured test data covering normal, boundary, negative, and exceptional scenarios while validating downstream business decisions, data integrity, and behavior across dependent application services.
- Logged and managed defects in Jira with reproducible steps, expected and actual results, supporting evidence, severity, and business impact while partnering with developers on root-cause analysis and fix validation.
- Integrated automated regression with **Jenkins, Git, and CI/CD pipelines**, supporting continuous testing, release validation, reusable test assets, and earlier detection of application and integration defects.
- Supported regulated enterprise delivery through test documentation, traceability, quality reporting, defect management, release evidence, and collaboration with technical and business stakeholders throughout Agile/Scrum delivery.

Environments: Java, Python, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, REST/SOAP APIs, JSON/XML, SQL, Jenkins, Git, Jira, CI/CD, Data Validation, Integration Testing, UAT, Traceability, Defect Management, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.